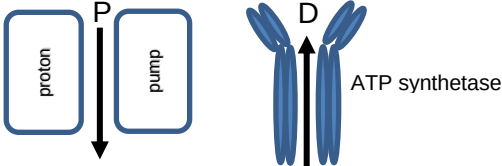


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)		Typical (in a layer) near {upper epidermis / adaxial surface} / most chloroplasts are in the palisade {layer / cells / mesophyll} / Spread across the surface (1) Millet concentrated around vascular bundle / owtte near both upper and lower epidermis / found spread throughout the leaf / owtte / ref to clusters (1)		2		2		2
		(ii)		{receive / are exposed to / absorb} light on both surfaces / neither side is in the shade(1) Chloroplasts are closer to the surface on both sides / less structures for light to pass through to reach chloroplasts / owtte (1)			2	2		
	(b)	(i)		Photolysis	1			1		
		(ii)		A_{II} Plotted At 1.8 above PSII, A_I plotted at 2.0 above PSI (1) Allow if points displaced horizontally but NOT vertically Tolerance = less than one small square Points must be labelled correctly Joined by three straight lines in correct pattern (1)		2		2	1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		Any five (x1) from A. In PS II {light / photons} {increase electron energy / excite electrons} (1) B. (higher energy electrons emitted and) accepted by {A_{II} / electron acceptor} / owtte (1) C. {Energy decreases / electrons lose energy} as pass through {ETC / electron carriers} (1) D. (Energy used to) {Pump protons / for ATP production / photophosphorylation} (1) E. PS I {electrons raised to higher energy level / excited} and accepted by {A_I / electron acceptor} (1) F. Use of data (1)	4	1		5		
(c)		(i)		 Both correct for 1 mark arrows must be labelled		1		1		
		(ii)		Any three (x1) from • Photolysis releases {H⁺ / protons} / photolysis increases concentration of protons} (1) • <u>inside thylakoid</u> {space / lumen} (1) • NADP→NADPH₂ / NADP {is reduced / decreases the concentration of protons} (1) • <u>in stroma</u> (1)		3		3		
Question 2 total					5	9	2	16	1	2